llmXive Automated Review of RynnWorld-4D: 4D Embodied World Models for Robotic Manipulation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper contains several critical factual inaccuracies regarding hardware, software versions, and data interpretation that undermine the validity of its central claims.

First, the hardware specification in Table 1 and Section 4.3 cites an "NVIDIA RTX 5090" GPU. As of the current date, this product does not exist in the public domain; the current flagship is the RTX 4090. Since the latency benchmarks (1.1s total cycle) are the primary evidence for the "real-time" feasibility of the system, the use of a non-existent hardware baseline renders these performance claims unverifiable and likely fabricated. This is a fatal flaw for a paper claiming real-time robotic control.

Second, the data curation pipeline relies on "Qwen3-VL" and "Depth Anything 3" (citations `bai2025qwen3`, `lin2025depth`). These model versions have not been publicly released as of the current knowledge cutoff (latest are Qwen2.5 and Depth Anything 2). If these models are internal or hypothetical, the claim that the dataset is "curated" using them is misleading. The authors must clarify if these are real, available models or if the dataset was generated using different, existing tools.

Third, there is a mathematical inconsistency in the control frequency claims. Table 1 reports a total inference latency of 1,106 ms (~0.9 Hz). The text claims an "effective control frequency of ~9 Hz" by executing 10 actions per cycle. However, the text also states the robot executes the *previously* planned chunk while the *next* cycle computes. If the computation takes 1.1s, the update rate is 0.9 Hz. The 9 Hz figure implies the robot is acting at 50 Hz *between* updates, but the "effective control frequency" usually refers to the policy update rate in this context. The discrepancy between the 1.1s bottleneck and the 9Hz claim needs rigorous clarification or correction.

Finally, the results interpretation in Section 4.3 contradicts Table 1. The text claims the proposed model "significantly outperforming" baselines like Wan-2.1-I2V-14B. However, the table shows Wan-2.1-I2V-14B achieves a higher Imaging Quality (IQ) score (0.684 vs 0.635) and Motion Smoothness (MS) (0.988 vs 0.995 is close, but IQ is lower). The claim of superiority is not supported by the reported metrics for these specific dimensions.

Action items.

- [**fatal**] The paper contains several critical factual inaccuracies regarding hardware, software versions, and data interpretation that undermine the validity of its central claims. First, the hardware specification in Table 1 and Section 4.3 cites an "NVIDIA RTX 5090" GPU. As of the current date, this product does not exist in the public domain; the current flagship is the RTX 4090. Since the latency benchmarks (1.1s total cycle) are the primary evidence for the "real-time" feasibility of the system, the

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`
Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure effectively visualizes the model's pipeline and outputs, but the caption contains grammatical errors and a raw citation placeholder that should be corrected.

Figure 2

The figure effectively visualizes the dataset composition with clear donut charts, but the bottom portion of the image is cropped, cutting off the legend and title, and some text labels are too small to read clearly.

Figure 3

The figure effectively illustrates the data curation pipeline and aligns with the caption's description of the annotation process. However, the legibility is compromised by text overlaid on the image grid and vertically rotated labels in the central flowchart.

Figure 4

The figure provides a clear visual overview of the pipeline architecture, but the caption is critically flawed with missing model names and incomplete sentences, preventing a full understanding of the system described.

Figure 5

Figure 5 effectively illustrates the six diverse tasks comprising the real-world manipulation benchmark described in the caption. The visual layout is clear, with distinct panels for each task and numbered arrows indicating motion sequences, requiring no additional legends or axes.

Figure 6

The figure effectively displays qualitative results with clear visual separation between modalities, but the caption contains a missing subject placeholder and fails to explicitly label the fourth column shown in the grid.

Figure 7

Figure 7 effectively illustrates the standardized experimental platform described in the caption, clearly displaying the TIANJI M6 robotic arm and WUJI Hand with their respective specifications (DOF, weight, force) labeled directly on the image. The visual presentation is clean, legible, and fully supports the caption's claim of a unified hardware configuration.

Figure 8

The figure is severely compromised by a large graphic obscuring the subject, and the caption is grammatically incomplete and fails to describe the data collection context.

Figure 9

The figure provides qualitative results but fails to display the depth maps described in the caption, and the caption text contains a missing model name.

Action items.

- [**writing**] Figure 1: The caption contains a grammatical error ('generates RGB...') missing the subject (e.g., 'the model' or 'RynnWorld-4D'), making it unclear what performs the action.
- [**writing**] Figure 1: The caption includes a citation placeholder '[teasor.pdf]' which should be removed or replaced with the proper citation format.
- [**writing**] Figure 2: The title 'Rynn4DDataset 1.0' and the bottom legend bar are cut off at the bottom edge of the image, making the full legend and total frame count illegible.
- [**writing**] Figure 2: The text labels for the 'Human demonstration videos' pie chart (e.g., 'EgoVid', 'Epic-Kitchens') are extremely small and blurry, reducing readability.
- [**writing**] Figure 3: The text 'Rynn4DDataset 1.0' is superimposed directly over the grid of sample images, significantly reducing the legibility of the underlying visual content.
- [**writing**] Figure 3: The text inside the vertical annotation boxes ('Video captioning', 'Flow annotation', 'Depth annotation') is rotated 90 degrees, making it difficult to read compared to standard horizontal text.
- [**fatal**] Figure 4: The caption contains multiple missing model names (e.g., 'Overview of .', 'co-generates', 'aggregated by -Policy'), rendering the text description incomplete and unprofessional.
- [**science**] Figure 4: The diagram shows 'RynnWorld-4D Block' and 'RynnWorld-4D-Policy' as distinct components, but the caption fails to name the generative model, making it impossible to verify if the visual architecture matches the described system.
- [**writing**] Figure 4: The text 'RynnWorld-4D Block' and 'RynnWorld-4D-Policy' are used in the diagram, but the caption does not define these terms or explain their relationship to the overall pipeline.
- [**writing**] Figure 6: The caption contains a grammatical error and missing subject ('Qualitative results of . Starting from...'), likely due to a missing model name placeholder.
- [**writing**] Figure 6: The caption claims to show 'RGB video, depth maps, and optical flow', but the figure displays four columns per task (RGB, Depth, Flow, and a fourth unlabelled column) without explicitly defining the fourth column in the text.

- **[writing]** Figure 8: The caption 'Operator to collect real world data' is grammatically incomplete and lacks context; it should describe the operator's role or the specific data collection setup shown.
- **[science]** Figure 8: The image contains a large, opaque smiley-face graphic obscuring the operator's head and upper torso, which prevents verification of the operator's identity, safety gear, or interaction with the equipment.
- **[writing]** Figure 9: The caption contains a grammatical error ('highlight 's ability') where the model name is missing.
- **[science]** Figure 9: The figure displays only RGB and optical flow sequences but lacks the depth maps explicitly promised in the caption ('Each row displays the generated RGB, depth, and optical flow sequences').

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-written for a specialized audience, but it contains several instances of undefined notation and in-group shorthand that would stall a competent reader from an adjacent field (e.g., a computer vision researcher not specializing in diffusion transformers or a robotics researcher new to 4D generation).

The most significant issue is the introduction of the threshold parameter τ in Section 3.1 (Eq. 1) without definition. The text states "masking out pixels where $\|\nabla D\| > \tau$" but never specifies what τ represents, its value, or its units. This forces the reader to guess or search the appendix, breaking the flow of understanding the geometric reconstruction pipeline.

Additionally, the paper relies on specific architectural shorthand. "3D RoPE" is used frequently in Section 3.2 without being expanded to "3D Rotary Positional Embeddings" or explained as a spatial-temporal variant of the standard RoPE. While "RoPE" is standard in LLMs, the "3D" extension is a specific modification here that warrants a brief gloss. Similarly, Table 1 uses "bb." for "backbone," which is lab slang that should be expanded to "backbone" for formal publication.

Finally, the metrics in Section 4.2 (IQ, MS, SC, Subj.) are introduced as a list of acronyms with a pointer to the appendix. For a self-contained reading experience, these should be expanded at their first occurrence in the main text (e.g., "Imaging Quality (IQ)"), allowing the reader to understand the evaluation axes immediately without navigating to the appendix.

These are minor, text-only fixes that significantly improve accessibility for the target "adjacent-field PhD" audience.

Action items.

- **[writing]** Section 3.1 (Eq. 1) introduces the symbol au in the phrase 'masking out pixels where $\| abla D\| > au$‘ without defining it. A competent reader cannot determine the threshold value or its units. Define au explicitly (e.g., ‘where au is a depth-gradient threshold set to 0.1’).

- [**writing**] Section 3.2 introduces the term ‘3D RoPE’ (3D Rotary Positional Embeddings) without expansion or a brief gloss. While ‘RoPE’ is common in NLP, the ‘3D’ variant (spatial + temporal) is specific to this architecture. Expand at first use: ‘3D Rotary Positional Embeddings (3D RoPE), which inject spatial and temporal position information’.
- [**writing**] Section 3.2 (Eq. 3) uses the notation $mK_l^{extcross}$ and $mV_l^{extcross}$ without explicitly defining the concatenation operation or the set of indices $jeqm$ in the immediate text. While the equation implies it, a reader might miss that ‘cross’ refers to the concatenation of the *other two* modalities. Add a clause: ‘where $mK_l^{extcross}$ is the concatenation of keys from the two complementary modalities ($jeqm$)’.
- [**writing**] Section 4.1 (Implementation Details) uses the abbreviation ‘bb.’ in ‘joint (frozen bb.)’ in Table 1. This is in-group shorthand for ‘backbone’ and is undefined. Expand to ‘joint (frozen backbone)’ for clarity.
- [**writing**] Section 4.2 (Setups and Baselines) lists metrics ‘IQ, MS, SC, Subj.’ without defining them in the main text, referring readers to the Appendix. While the Appendix defines them, the main text should briefly expand these acronyms at first mention (e.g., ‘Imaging Quality (IQ), Motion Smoothness (MS), Subject Consistency (SC), and I2V-Subject (Subj.)’) to allow immediate comprehension without page-flipping.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a logically coherent argument for using synchronized RGB, depth, and optical flow (RGB-DF) to bridge 2D video generation and 3D robotic control. The progression from identifying 2D limitations to proposing the RGB-DF representation and validating it via ablation studies is sound. The internal logic of the architecture (tri-branch design, joint attention) is well-supported by the experimental results.

However, there are specific inconsistencies in numerical claims and terminology that disrupt the logical flow:

1. **Control Frequency Contradiction:** Section 4.1 states the system executes 10 actions at 50 Hz during a 1.1s inference cycle. Mathematically, 10 actions over 1.1 seconds yields ~9 Hz, not 50 Hz. The text conflates the low-level servo loop rate (500 Hz) with the action execution rate, creating a logical gap between the stated cycle time and the claimed execution frequency.

2. **Abstract vs. Body Mismatch:** The Abstract describes the control as "high-frequency," while the body (Sections 3.4 and 5) explicitly limits the system to 9 Hz. In robotics, "high-frequency" typically implies >50-100 Hz. This creates a misleading impression that contradicts the specific limitations admitted later in the paper.
3. **Depth Clipping and Metric Validity:** The method clips depth to [0.0, 5.0] meters (Section 3.2) but claims to derive "metric scene flow" for "physically plausible" movements (Section 3.3). If the environment extends beyond 5.0 meters, the clipping introduces a geometric discontinuity that invalidates the "metric" nature of the reconstruction for distant objects. The paper does not reconcile this constraint with its claims of physical plausibility in general scenes.

Addressing these points will ensure the paper's conclusions strictly follow from its stated premises and data.

Action items.

- [**writing**] Section 4.1 claims 50 Hz execution during a 1.1s cycle for 10 actions, which mathematically implies ~9 Hz. Clarify if 50 Hz refers to the low-level servo holding a static command, as the numbers 9 Hz, 50 Hz, and 1.1s are currently inconsistent.
- [**writing**] The Abstract claims 'high-frequency' control, but Section 3.4 and Conclusion admit a 9 Hz limit. This contradicts standard robotics definitions of high-frequency (>50Hz). Qualify 'high-frequency' in the Abstract to align with the 9 Hz finding.
- [**science**] Section 3.2 clips depth to [0, 5.0]m, yet Section 3.3 claims 'metric scene flow' for 'physically plausible' 3D movements. If scenes exceed 5m, clipping flattens geometry, invalidating the metric claim. Explain how this constraint affects 3D validity in large scenes.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims in its Abstract and Conclusion that exceed the scope of the experimental evidence provided. Specifically, the Abstract asserts that the method "solves" the gap between world prediction and policy learning and achieves "state-of-the-art performance" on real-world dexterous manipulation. However, the results in Section 4 (Table 2) show that while RynnWorld-Policy outperforms baselines on tasks requiring high precision (e.g., Lid Placement, Hand-over), it does not achieve SOTA on all tasks; for instance, on "Block Pushing," the $\pi_{0.5}$ baseline achieves 100% success compared to RynnWorld-Policy's 97.14%. The use of absolute terms like "solves" and unqualified "state-of-the-art" overstates the findings, which are currently limited to a specific subset of tasks and a single robot platform (TIANJI M6).

Furthermore, the Conclusion frames the work as a "paradigm shift" for "general-purpose embodied intelligence.” While the tri-branch architecture and 4D representation are novel and effective for the tested scenarios, the evidence is restricted to egocentric views and six specific manipulation tasks. The paper does not demonstrate generalization to multi-view systems, different robot morphologies, or truly open-world environments, which the “open world” language in the Introduction suggests. The “Limitation” section correctly identifies the egocentric constraint, but this critical boundary is not reflected in the confident, universal language of the Abstract and Conclusion.

To align the rhetoric with the evidence, the authors should:

1. Replace “solves” with “significantly narrows” or “addresses” in the Abstract.
2. Qualify the “state-of-the-art” claim to specify the tasks where it holds true or acknowledge the exceptions.
3. Temper the “paradigm shift” language in the Conclusion to reflect that this is a promising step rather than a completed revolution, given the narrow experimental scope.
4. Ensure the Abstract explicitly mentions the egocentric limitation to prevent readers from assuming broader applicability than demonstrated.

Action items.

- [**writing**] Abstract claims the method ‘solves’ the gap between world prediction and policy learning. Section 4 shows a 34% relative improvement on specific bimanual tasks (Hand-over, Lid Placement) but fails on others (e.g., Block Pushing is 97.14% vs 100% baseline). Replace ‘solves’ with ‘significantly narrows’ and scope the claim to the tested manipulation tasks.
- [**writing**] Abstract states the model achieves ‘state-of-the-art performance’ on ‘real-world dexterous bimanual manipulation tasks.’ Table 2 shows SOTA only on 4 of 6 tasks (Hand-over, Lid Placement, Bowl Stacking, Bimanual Lifting) and is outperformed by pi_0.5 on Block Pushing (100% vs 97.14%). Qualify the claim to ‘on tasks requiring high spatial precision’ or list the specific tasks where SOTA was achieved.
- [**writing**] Conclusion states the framework ‘shifts the paradigm’ from 2D to 4D. While the results show improvement, the evidence is limited to a single robot platform (TIANJI M6) and six specific tasks. The claim of a paradigm shift implies a broader field-wide impact not yet demonstrated. Rephrase to ‘demonstrates a promising shift’ or ‘offers a new paradigm for’ to reflect the preliminary nature of the evidence.
- [**writing**] The ‘Limitation’ section admits the model is ‘primarily optimized for egocentric perspectives’ but the Introduction and Abstract imply general applicability to ‘open world’ and ‘complex 3D world’ interactions. The paper does not test multi-view or non-egocentric scenarios. Explicitly state in the Abstract that the current validation is restricted to egocentric views to prevent overgeneralization.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a generative world model for robotic manipulation and a large-scale dataset (Rynn4DDataset 1.0) derived from public sources. From a safety and ethics perspective, the work is low-risk.

The dataset construction relies on publicly available video datasets (e.g., Epic-Kitchens, EgoVid, RoboMIND) and applies automated pseudo-labeling (depth, flow, captions) using existing models. The paper does not claim to release raw, unprocessed video containing Personally Identifiable Information (PII) or sensitive human data; rather, it releases the processed dataset or the pipeline to generate it. While the source videos may contain human faces or activities, the paper does not describe a novel method for re-identifying individuals or scraping data in violation of Terms of Service (ToS) beyond standard academic usage of public benchmarks. The authors cite the source datasets, implying adherence to their respective licenses.

The proposed system (RynnWorld-4D and RynnWorld-4D-Policy) is designed for physical robotic manipulation in controlled or semi-controlled environments. It does not introduce dual-use capabilities for surveillance, deception, or cyber-attacks. The "world model" aspect predicts physical dynamics (RGB, depth, flow) to aid robot control, which is a standard and benign application in embodied AI. There is no evidence of human-subjects research requiring IRB approval, as the data is secondary use of public datasets and the robot experiments involve teleoperation by the authors (standard engineering practice) rather than behavioral studies on human subjects.

No specific, non-trivial risks of harm are identified that are unaddressed in the text. The paper does not require additional safety disclosures or mitigations beyond standard academic practice for this subfield.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling architecture for 4D world modeling, but the experimental design contains several gaps that prevent the results from definitively supporting the strength of the claims made.

First, the ablation studies in Table 2 (World Modeling) and Table 3 (Policy Learning) rely on single-point estimates without reporting variance. For instance, the improvement in geometric accuracy (1) when adding Modality Adaptation is 0.131 (0.610 vs 0.479). In generative modeling, where results can fluctuate significantly based on random seeds and initialization, a single-run comparison is insufficient to rule out luck. The authors must report mean and standard deviation across at least 3-5 independent training runs for all ablation variants to demonstrate that the observed gains are stable and not artifacts of a specific random seed.

Second, the comparison against foundation models like 0.5 in Table 3 raises concerns about baseline fairness. The baseline achieves 0.00% success on the “Hand-over” task, which is an extreme failure rate for a 35-trial set. This suggests the baseline may have been under-tuned for the specific dexterous hand configuration (WUJI HAND) or the task setup, rather than failing due to the inherent limitations of 2D representations. The paper does not disclose whether the baselines underwent a hyperparameter search comparable to the proposed method. To support the claim that 4D representations are “essential,” the authors must ensure baselines are given a fair chance to succeed, either by reporting tuned results or by increasing the trial count to reduce the impact of stochastic failure.

Third, the “w/o 4D Pre-training” ablation claims that the performance collapse is due to data scarcity. However, the text does not explicitly confirm that this ablation was trained for the same number of steps or epochs as the full model. If the ablation was trained for fewer iterations, the poor performance could be attributed to under-training rather than the lack of large-scale data. The authors need to explicitly state that compute budgets were held constant across all ablation experiments.

Finally, the latency breakdown in Table 1 includes “Depth Estimation (DA3)” as a 7.7% overhead. Since the proposed model *generates* depth, the inclusion of an external depth estimator (Depth Anything 3) in the inference pipeline is confusing. If this step is only for ground-truth generation during evaluation, it should not be in the “Inference Latency” table for the policy. If it is part of the deployed system, the authors must clarify why an external estimator is needed when the world model already predicts depth, and ensure the reported 9 Hz frequency accounts for this correctly.

Action items.

- [**science**] The paper presents a compelling architecture for 4D world modeling, but the experimental design contains several gaps that prevent the results from definitively supporting the strength of the claims made. First, the ablation studies in Table 2 (World Modeling) and Table 3 (Policy Learning) rely on single-point estimates without reporting variance. For instance, the improvement in geometric accuracy (1) when adding Modality Adaptation is 0.131 (0.610 vs 0.479). In generative modeling, where resu

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in the results section requires minor revisions to align with standard inferential practices. While the experimental design (35 trials per task) is sufficient for estimation, the presentation of results lacks necessary uncertainty quantification and rigorous hypothesis testing.

First, **Table 2** (Success rates) presents point estimates to two decimal places (e.g., 94.29%) derived from exactly 35 trials. This implies a precision that the sample size cannot support (the standard error for a 94% success rate with N=35 is approximately 3.5%). More critically, the table lacks any measure of variance (standard deviation, standard error, or confidence intervals). Without these, a reader cannot determine if the observed differences between methods (e.g., 94.29% vs. 91.43%) are statistically meaningful or within the noise of the sampling distribution. The authors should report mean ± SD or 95% confidence intervals for all success rates.

Second, **Table 1** involves a large number of pairwise comparisons (4 baselines × 10 metrics = 40 tests). The paper highlights the "best" performing method in bold for each metric without applying any correction for multiple comparisons (such as Bonferroni, Holm, or Benjamini-Hochberg). In a set of 40 independent tests at =0.05, one would expect approximately 2 false positives purely by chance. The current presentation risks overstating the significance of the proposed method's advantages. The authors should either apply a multiple-comparison correction to the p-values (if tests were run) or explicitly clarify that the bolding indicates the highest numerical value, not a statistically significant difference.

Finally, the text in **Section 4.2** repeatedly uses the term "significantly outperforming" or "significantly better" (e.g., regarding Lid Placement and Bowl Stacking) without referencing a specific statistical test, p-value, or effect size. In scientific writing, "significant" should strictly refer to a result that has passed a hypothesis test. If no formal test was conducted, the language should be softened to "numerically higher" or "improved," and the uncertainty metrics mentioned above should be provided to support the magnitude of the improvement.

Action items.

- [**writing**] Table 2 reports success rates (e.g., 94.29%) for 6 tasks based on 35 trials each, but provides no uncertainty metrics (SD, SE, or 95% CI). With N=35, the standard error for a 94% rate is ~3.5%, making the difference between 94.29% and 91.43% statistically indistinguishable without a formal test. Report mean ± SD or 95% CIs for all success rates and clarify if the bolding implies statistical significance.
- [**writing**] Table 1 compares the proposed method against 4 baselines across 10 metrics (40 pairwise comparisons) and highlights 'best' values in bold without any multiple-comparison correction (e.g., Bonferroni or FDR). At =0.05, ~2 false positives are expected by chance. Apply a correction across the 40 tests or explicitly state that the bolding is for visual ranking only, not statistical significance.
- [**writing**] Section 4.2 claims the method is 'significantly outperforming' baselines in specific tasks (e.g., Lid Placement) but cites no p-values, effect sizes, or statistical tests (e.g., Fisher's exact test or chi-square for success rates). Replace 'significantly' with 'numerically higher' unless a formal hypothesis test with p-values is added to the text or appendix.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured and the technical narrative is strong, but there are several instances where the prose flow is interrupted by formatting choices or abrupt transitions that force the reader to re-parse the intent.

In Section 3.1, the "Video captioning" paragraph breaks the narrative rhythm by inserting a raw prompt block immediately after a colon. This creates a visual and logical gap where the reader expects a continuation of the sentence or a clear explanation of the prompt's role before seeing the code. Integrating the prompt's purpose into the sentence or moving the block to a figure would smooth this transition.

Section 3.2 suffers from a similar issue in the "Metric Scene Flow Derivation" paragraph. The introduction of the tracking equation is abrupt, lacking a clear verbal bridge from the concept of "temporal correspondences" to the specific mathematical operation. A simple connecting phrase would clarify the logical progression.

In Section 3.3, the description of the training stages assumes too much continuity. The transition from Stage 2 to Stage 3 mentions the "joint module" without explicitly naming the "Joint Cross-Modal Attention" modules again, which may cause a slight cognitive load for the reader trying to track the specific components being unfrozen.

Section 3.4 and 3.5 have a structural misalignment. The "Action Generation" paragraph refers the reader to the latency section for details on "parallel action chunking," but the latency section focuses primarily on timing breakdowns. This forces the reader to search for the actual mechanism description, disrupting the flow of understanding the policy's operation.

Finally, in Section 4.2, the list of robot tasks is mostly consistent in style, but the "Bowl Stacking" item deviates by starting with a narrative scene setting ("There are two small bowls...") rather than a direct task description. Aligning this with the style of the other items would improve the section's professional polish.

These issues are minor and do not obscure the core scientific contributions, but addressing them would significantly improve the reader's ability to move through the text without friction.

Action items.

- **[writing]** Section 3.1: The paragraph on 'Video captioning' opens with a prompt block that interrupts the narrative flow. The sentence 'For each segment, we provide the following prompt to the model:' is followed immediately by a quote block, leaving the reader hanging. Integrate the prompt description into the sentence or move the block to a figure caption to maintain prose continuity.
- **[writing]** Section 3.2: The paragraph on 'Metric Scene Flow Derivation' contains a sentence ('A 3D point P_t is tracked to its position at t+1 by:') that introduces an equation but lacks a clear subject-verb connection to the preceding context. The transition from the previous

sentence about 'temporal correspondences' to the specific tracking formula is abrupt. Add a bridging phrase like 'Specifically, we track a point by...' to clarify the logical step.

- [**writing**] Section 3.3: The 'Phased Training Strategy' subsection lists three stages but the transition between Stage 2 and Stage 3 descriptions is abrupt. The sentence 'With the joint module already aligned, we unfreeze the entire model...' assumes the reader remembers the specific alignment mechanism from Stage 2. Add a brief reference to the 'Joint Cross-Modal Attention' modules to reinforce the continuity of the argument.
- [**writing**] Section 3.4: The 'Action Generation' paragraph ends with a reference to 'Sec.~ ef{appendix:latency}' for details on parallel action chunking, but the latency section (3.5) is titled 'Inference Latency and Real-time Control' and focuses on timing breakdown rather than the chunking mechanism itself. The reader must hunt for the chunking explanation. Either rename the section to include 'Action Chunking' or move the chunking explanation to the 'Action Generation' paragraph.
- [**writing**] Section 4.2: The 'Robot tasks' list item (6) 'Bowl Stacking' starts with 'There are two small bowls on the table...', which is a narrative description rather than a task definition. The other items use imperative or descriptive task structures (e.g., 'The robot uses...', 'A sequential pushing task...'). Rewrite this item to match the consistent task-definition style of the other list items.